

From Spin- $\frac{1}{2}$ to the Entangled Readout of the Proton Spin

A study companion to the Tessera emergent-proton construction

Tessera theory notes (part of issue #410)

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Abstract

This is a self-contained study document. It walks, rung by rung, from the quantum mechanics of a single spin- $\frac{1}{2}$ particle up to the central problem solved (and partly still open) in the Tessera codebase: reading the *composite* total spin of the emergent proton bound state and recovering the value $J^2 = \frac{3}{4}$. Every symbol is defined where it first appears. The physics ladder (Sections 1–4) is standard and can be checked against any quantum-mechanics or particle-physics textbook; the geometry and Dirac-Kähler material (Sections 5–6) sets up the discrete language the code speaks; the last three sections (Sections 7–10) connect that language to the actual modules and findings. The pivotal conceptual point, established in Section 3 and paid off in Sections 8–9, is that the proton’s spin is an *entangled* three-body state, and that any readout which reduces each quark to its own qubit *provably* cannot reproduce $\frac{3}{4}$.

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1 One spin- $\frac{1}{2}$: the smallest quantum top

A spin- $\frac{1}{2}$ particle is the simplest non-trivial quantum system: a two-dimensional complex Hilbert space $\mathcal{H} = \mathbb{C}^2$. We pick an orthonormal basis and name the basis vectors

$$|\uparrow\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |\downarrow\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad (1)$$

“spin up” and “spin down” along a chosen quantization axis (conventionally $\hat{z}$). A general (pure) state is a normalized complex superposition $|\psi\rangle = \alpha |\uparrow\rangle + \beta |\downarrow\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$.

The Pauli matrices. The three Hermitian, traceless 2×2 Pauli matrices are

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (2)$$

They satisfy

$$\sigma_a \sigma_b = \delta_{ab} \mathbb{I} + i \epsilon_{abc} \sigma_c, \quad (3)$$

where δ_{ab} is the Kronecker delta, ϵ_{abc} is the totally antisymmetric Levi-Civita symbol ($\epsilon_{123} = +1$), $\mathbb{I}$ is the 2×2 identity, and repeated indices are summed (Einstein convention). Equation (3) packages two facts: the *anticommutator* $\{\sigma_a, \sigma_b\} = 2\delta_{ab}\mathbb{I}$ (so each $\sigma_a^2 = \mathbb{I}$), and the *commutator* $[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c$.

The spin operators. The physical spin observables (in units $\hbar = 1$, used throughout) are

$$S_a := \frac{1}{2} \sigma_a, \quad a \in \{1, 2, 3\}. \quad (4)$$

From $[\sigma_a, \sigma_b] = 2i\epsilon_{abc}\sigma_c$ we get the defining angular-momentum algebra

$$[S_a, S_b] = i\epsilon_{abc} S_c. \quad (5)$$

This is the Lie algebra $\mathfrak{su}(2)$; any set of three operators obeying (5) is a representation of spin/angular momentum, a fact we will use repeatedly.

The Casimir. The total-spin-squared operator

$$S^2 := S_1^2 + S_2^2 + S_3^2 \quad (6)$$

commutes with every S_a (it is the *Casimir* of $\mathfrak{su}(2)$). Using $\sigma_a^2 = \mathbb{I}$,

$$S^2 = \frac{1}{4}(\sigma_1^2 + \sigma_2^2 + \sigma_3^2) = \frac{1}{4} \cdot 3\mathbb{I} = \frac{3}{4}\mathbb{I}. \quad (7)$$

Comparing with the general eigenvalue $s(s+1)$ of S^2 gives $s(s+1) = \frac{3}{4} \Rightarrow s = \frac{1}{2}$: every state of $\mathbb{C}^2$ is a spin- $\frac{1}{2}$ state, and $\frac{3}{4}$ is the fingerprint of a single spin- $\frac{1}{2}$ degree of freedom. **Keep the number $\frac{3}{4}$ in view:** it is the value the proton must also reproduce, for a quite different (composite) reason.

Measurement and expectation values. A measurement of S_a yields one of its eigenvalues $\pm\frac{1}{2}$, and in the state $|\psi\rangle$ the average (expectation value) over many measurements is

$$\langle S_a \rangle := \langle \psi | S_a | \psi \rangle. \quad (8)$$

The triple $\langle \mathbf{S} \rangle = (\langle S_1 \rangle, \langle S_2 \rangle, \langle S_3 \rangle)$ is the *Bloch vector*; for a pure state it has length $|\langle \mathbf{S} \rangle| = \frac{1}{2}$. Equivalently the state’s density matrix is $\rho = \frac{1}{2}(\mathbb{I} + 2\langle \mathbf{S} \rangle \cdot \boldsymbol{\sigma})$. This Bloch picture is exactly how the code will summarize the spin extracted at each “hole” in Section 7, and the fact that a Bloch vector keeps *only* $\langle \mathbf{S} \rangle$ (three real numbers) is precisely what loses the entanglement in Section 8.

2 Two spins: the singlet/triplet split $2 \otimes 2 = 1 \oplus 3$

Tensor products. Two particles live in the tensor-product space $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^4$, spanned by $|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle$, where $|ab\rangle := |a\rangle \otimes |b\rangle$. An operator acting on particle 1 alone is $A \otimes \mathbb{I}$; on particle 2 alone, $\mathbb{I} \otimes B$. The *total* spin operator is the sum

$$\mathbf{S}_{\text{tot}} = \mathbf{S}^{(1)} \otimes \mathbb{I} + \mathbb{I} \otimes \mathbf{S}^{(2)}, \quad (9)$$

which still obeys (5) because the two single-particle algebras commute with each other.

Total Casimir. With $\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} := \sum_a S_a^{(1)} S_a^{(2)}$,

$$S_{\text{tot}}^2 = (\mathbf{S}^{(1)})^2 + (\mathbf{S}^{(2)})^2 + 2 \mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} = \frac{3}{4} + \frac{3}{4} + 2 \mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}. \quad (10)$$

The cross term $\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)}$ is what couples the two spins; its eigenvalues sort the four states into multiplets.

Clebsch–Gordan: the decomposition. The representation-theory statement is

$$\boxed{2 \otimes 2 = 1 \oplus 3} \quad (11)$$

i.e. $\mathbb{C}^2 \otimes \mathbb{C}^2$ splits into a one-dimensional *singlet* ($S = 0$) and a three-dimensional *triplet* ($S = 1$). Explicitly, in terms of the total magnetic quantum number m (the S_{tot}^z eigenvalue):

$$\begin{aligned} \text{triplet } (S = 1) : \quad & |1, +1\rangle = |\uparrow\uparrow\rangle, \\ & |1, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \\ & |1, -1\rangle = |\downarrow\downarrow\rangle; \end{aligned} \quad (12)$$

$$\text{singlet } (S = 0) : \quad |0, 0\rangle = \frac{1}{\sqrt{2}} (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle). \quad (13)$$

The numerical prefactors $\pm \frac{1}{\sqrt{2}}$ are the *Clebsch–Gordan coefficients* for $\frac{1}{2} \otimes \frac{1}{2}$. Note the symmetry under exchange of the two particles: the triplet is *symmetric*, the singlet is *antisymmetric*. This symmetry bookkeeping is the seed of the Pauli argument in Section 4.

Eigenvalues. On the triplet, $S_{\text{tot}}^2 = 1(1+1) = 2$; on the singlet, $S_{\text{tot}}^2 = 0(0+1) = 0$. Cross-checking with (10): $\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} = +\frac{1}{4}$ on the triplet and $-\frac{3}{4}$ on the singlet, so $S_{\text{tot}}^2 = \frac{3}{2} + 2(\frac{1}{4}) = 2$ and $\frac{3}{2} + 2(-\frac{3}{4}) = 0$ respectively. Already here we see that S_{tot}^2 is *not* a property of either particle alone: it is fixed by the relative phase/sign in (12)–(13).

3 Three spins: $2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2$ and the entangled proton

Now combine three spin- $\frac{1}{2}$ particles, the case directly relevant to a baryon (three quarks). The Hilbert space is $\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2 \cong \mathbb{C}^8$, with total spin operator $\mathbf{J} := \mathbf{S}^{(1)} + \mathbf{S}^{(2)} + \mathbf{S}^{(3)}$ (we switch to $\mathbf{J}$ for the three-body total, to match the baryon literature). We use $|abc\rangle$ for $|a\rangle \otimes |b\rangle \otimes |c\rangle$, and write e.g. $|uud\rangle$ when we later attach flavor labels u, d ; here read $u \equiv \uparrow$, $d \equiv \downarrow$ as spin states.

The decomposition. Building up $2 \otimes 2 \otimes 2 = (1 \oplus 3) \otimes 2 = 2 \oplus (4 \oplus 2)$,

$$\boxed{2 \otimes 2 \otimes 2 = 4 \oplus 2 \oplus 2} \quad (14)$$

a four-dimensional *quartet* with $J = \frac{3}{2}$ and *two* two-dimensional *doublets* with $J = \frac{1}{2}$. The corresponding Casimir $J^2 = j(j+1)$ values are

$$J = \frac{3}{2} : \quad J^2 = \frac{3}{2} \cdot \frac{5}{2} = \frac{15}{4}, \quad J = \frac{1}{2} : \quad J^2 = \frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}. \quad (15)$$

The quartet ($\frac{15}{4}$) is the Δ baryon's spin; the doublets ($\frac{3}{4}$) are the proton/neutron's. The two doublets are distinguished by their behaviour under permutations of the three particles: they are *mixed-symmetry* representations (neither fully symmetric nor fully antisymmetric), conventionally split into a piece symmetric under $1 \leftrightarrow 2$ and a piece antisymmetric under $1 \leftrightarrow 2$. The quartet is *fully symmetric*.

The fully-symmetric quartet. The top of the $J = \frac{3}{2}$ tower is $|\frac{3}{2}, \frac{3}{2}\rangle = |\uparrow\uparrow\uparrow\rangle (= |uuu\rangle$ with all spins up). It is manifestly symmetric under any exchange of the three particles, and one checks directly $J^2 |uuu\rangle = \frac{15}{4} |uuu\rangle$.

The mixed-symmetry $J = \frac{1}{2}$ “proton” state. The piece of $J = \frac{1}{2}$, $m = +\frac{1}{2}$ that is *symmetric* under exchanging particles $1 \leftrightarrow 2$ is, up to normalization,

$$|\chi^{\text{MS}}\rangle \propto 2|\uparrow\uparrow\downarrow\rangle - |\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle = 2|uud\rangle - |udu\rangle - |duu\rangle. \quad (16)$$

(Here the labels u, d are spin up/down; the same combinatorial object appears again in Section 4 with u, d reading as *flavor*.) The normalized state is $|\chi\rangle = \frac{1}{\sqrt{6}}(2|uud\rangle - |udu\rangle - |duu\rangle)$.

Let us **verify** $J^2 |\chi\rangle = \frac{3}{4} |\chi\rangle$. Write $J^2 = \sum_i (\mathbf{S}^{(i)})^2 + 2 \sum_{i < j} \mathbf{S}^{(i)} \cdot \mathbf{S}^{(j)} = \frac{9}{4} \mathbb{I} + 2(\mathbf{S}^{(1)} \cdot \mathbf{S}^{(2)} + \mathbf{S}^{(1)} \cdot \mathbf{S}^{(3)} + \mathbf{S}^{(2)} \cdot \mathbf{S}^{(3)})$. The convenient identity for any two spin- $\frac{1}{2}$'s is

$$\mathbf{S}^{(i)} \cdot \mathbf{S}^{(j)} = \frac{1}{2} P_{ij} - \frac{1}{4} \mathbb{I}, \quad (17)$$

where P_{ij} is the *swap* operator exchanging particles i and j ($P_{ij} |\dots a_i \dots a_j \dots\rangle = |\dots a_j \dots a_i \dots\rangle$). Summing (17) over the three pairs,

$$J^2 = \frac{9}{4} \mathbb{I} + 2 \left[\frac{1}{2} (P_{12} + P_{13} + P_{23}) - \frac{3}{4} \mathbb{I} \right] = \frac{3}{4} \mathbb{I} + (P_{12} + P_{13} + P_{23}). \quad (18)$$

So we only need the action of the three swaps on $|\chi\rangle$. By construction (16) is symmetric in $1 \leftrightarrow 2$, so $P_{12} |\chi\rangle = + |\chi\rangle$. For the others, act on the unnormalized $|v\rangle = 2|uud\rangle - |udu\rangle - |duu\rangle$:

$$P_{13} |v\rangle = 2|duu\rangle - |udu\rangle - |uud\rangle, \quad (19)$$

$$P_{23} |v\rangle = 2|udu\rangle - |uud\rangle - |duu\rangle. \quad (20)$$

Adding all three swap results,

$$(P_{12} + P_{13} + P_{23}) |v\rangle = \underbrace{(2-1-1)}_{=0} |uud\rangle + \underbrace{(-1-1+2)}_{=0} |udu\rangle + \underbrace{(-1+2-1)}_{=0} |duu\rangle = 0. \quad (21)$$

Hence $(P_{12} + P_{13} + P_{23}) |\chi\rangle = 0$ and, from (18),

$$\boxed{J^2 |\chi\rangle = \frac{3}{4} |\chi\rangle}. \quad (22)$$

The state (16) is a genuine $J = \frac{1}{2}$ eigenstate. Notice it is an *entangled* superposition of three product kets with non-trivial relative coefficients $(2, -1, -1)$ — it cannot be written as a single $|a\rangle \otimes |b\rangle \otimes |c\rangle$.

The crucial counterexample: a plain product is *not* $\frac{3}{4}$. Contrast (16) with the bare product $|uud\rangle = |\uparrow\uparrow\downarrow\rangle$. Using (18): $P_{12}|uud\rangle = |uud\rangle$ (the first two are both $\uparrow$), while $P_{13}|uud\rangle = |duu\rangle$ and $P_{23}|uud\rangle = |udu\rangle$ are *different* kets. Therefore

$$J^2|uud\rangle = \frac{3}{4}|uud\rangle + |uud\rangle + |duu\rangle + |udu\rangle, \quad (23)$$

and the *expectation value* (the product ket is normalized, $\langle uud|uud\rangle = 1$, and is orthogonal to $|duu\rangle, |udu\rangle$) is

$$\boxed{\langle J^2 \rangle_{uud} = \langle uud|J^2|uud\rangle = \frac{3}{4} + 1 = \frac{7}{4}.} \quad (24)$$

A bare product of three spins gives $\frac{7}{4}$, *not* $\frac{3}{4}$. In fact $|uud\rangle = \frac{1}{\sqrt{6}}(\text{the } J=\frac{3}{2} \text{ part}) + \dots$ is a mixture of $J = \frac{3}{2}$ and $J = \frac{1}{2}$, and $\frac{7}{4}$ is the mixture-weighted average. *This is the single most important conceptual point of the whole document:* the proton’s $\frac{3}{4}$ lives in the relative $(2, -1, -1)$ phases — it is an *entanglement* property of the three-body state. Any procedure that treats the three quarks as independent (a product, or equivalently a per-particle Bloch reduction) cannot land on $\frac{3}{4}$; it floors at the mixture. We will meet exactly this floor in the lattice readout of Section 8.

4 Identical particles, exchange symmetry, and the baryon wavefunction

Why does Nature pick the specific entangled combination (16) for the proton, rather than $|uud\rangle$ or the quartet? The answer is the Pauli principle applied to the quarks’ *total* wavefunction.

Fermions and antisymmetry. Quarks are spin- $\frac{1}{2}$ fermions. The total wavefunction of a system of identical fermions must be *totally antisymmetric* under the simultaneous exchange of all labels of any two of them. A quark carries four kinds of label:

$$\Psi = \psi_{\text{space}} \otimes \chi_{\text{spin}} \otimes \phi_{\text{flavor}} \otimes \theta_{\text{color}}, \quad (25)$$

and exchange acts on all four factors at once. Antisymmetry of the product means the symmetries of the factors must *multiply* to “antisymmetric.”

A light touch of Young tableaux. The permutation group S_3 on three objects has three irreducible symmetry types, conveniently labelled by Young diagrams (written here as partitions):

$$\text{symmetric} \sim [3], \quad \text{mixed (MS, MA)} \sim [2, 1], \quad \text{antisymmetric} \sim [1, 1, 1].$$

(Pictorially: $[3]$ is a single row of three boxes, $[1, 1, 1]$ a single column of three, and $[2, 1]$ the “L”/hook shape.) In words: a single row = totally symmetric; a single column = totally antisymmetric; the “L”/hook shape = the two-dimensional *mixed-symmetry* representation, which comes as a pair MS, MA — symmetric resp. antisymmetric under the $1 \leftrightarrow 2$ exchange.) The combination rule we need is the *Clebsch–Gordan* of S_3 : the product of two mixed-symmetry doublets contains exactly one totally symmetric piece,

$$\text{MS} \otimes \text{MS} \supset \text{symmetric}, \quad (26)$$

formed as $\frac{1}{\sqrt{2}}(\phi^{\text{MS}}\chi^{\text{MS}} + \phi^{\text{MA}}\chi^{\text{MA}})$.

Color is the antisymmetric singlet. Each quark carries an $SU(3)_{\text{color}}$ index $a \in \{r, g, b\}$. A baryon is a color singlet, and the unique antisymmetric singlet of three color triplets is

$$\theta_{\text{color}} = \frac{1}{\sqrt{6}} \epsilon_{abc} |a\rangle |b\rangle |c\rangle, \quad (27)$$

with ϵ_{abc} the $SU(3)$ Levi–Civita symbol. This factor is *totally antisymmetric*. (In the Tessera construction, Section 7, this ϵ_{abc} singlet is realized not as an imposed index but as the *carried* representation $[1, \omega, \omega^2]$ on three emergent topological holes; $\omega = e^{2\pi i/3}$.)

The forcing argument. For the spatial ground state the three quarks sit in the lowest orbital, which is *symmetric*. So:

$$\underbrace{\Psi}_{\text{antisym}} = \underbrace{\psi_{\text{space}}}_{\text{sym}} \otimes \underbrace{(\phi_{\text{flavor}} \otimes \chi_{\text{spin}})}_{?} \otimes \underbrace{\theta_{\text{color}}}_{\text{antisym}} \Rightarrow \phi_{\text{flavor}} \otimes \chi_{\text{spin}} \text{ must be } \textit{symmetric}. \quad (28)$$

(antisym $\times$? $\times$ antisym $\times$ sym = antisym forces ? = sym). There are two ways to build a symmetric flavor–spin factor:

- (a) **symmetric flavor $\otimes$ symmetric spin** — both fully symmetric. Symmetric spin of three is the quartet, $J = \frac{3}{2}$. This is the Δ : e.g. $\Delta^{++} = |uuu\rangle$ with $J^2 = \frac{15}{4}$.
- (b) **mixed flavor $\otimes$ mixed spin**, combined into a symmetric whole via (26). Mixed spin of three is the doublet, $J = \frac{1}{2}$. This is the nucleon (proton/neutron), $J^2 = \frac{3}{4}$.

Why $\frac{3}{4}$ needs the 2+1 (uud) flavor structure. The proton has flavor content uud — *two* u quarks and *one* d . This 2+1 split is what *selects* the mixed–symmetry spin combination (16). Concretely: the two like quarks (the uu pair) sit in the flavor state symmetric under their exchange, and to make the total flavor–spin factor symmetric (26) the uu pair’s *spin* must also be symmetric, i.e. in the spin–triplet $S_{uu} = 1$. The full $J = \frac{1}{2}$ proton spin–flavor wavefunction is the $SU(6)$ state

$$|p\uparrow\rangle = \frac{1}{\sqrt{18}} (2|u\uparrow u\uparrow d\downarrow\rangle - |u\uparrow u\downarrow d\uparrow\rangle - |u\downarrow u\uparrow d\uparrow\rangle + (\text{permutations of which quark is } d)), \quad (29)$$

whose spin part, at fixed flavor ordering uud , is exactly the $2|\uparrow\uparrow\downarrow\rangle - |\uparrow\downarrow\uparrow\rangle - |\downarrow\uparrow\uparrow\rangle$ of (16). **The upshot for the code:** you cannot identify the $J = \frac{1}{2}$ combination, and hence the value $\frac{3}{4}$, without keeping the flavor labels and the spin labels *jointly*; if you sum spins blindly over three indistinguishable holes you land on the symmetric/quartet or the product mixture, not the mixed–symmetry proton. This is the physics reason the codebase must read flavor and spin *together* (Sections 8–9).

5 A primer on discrete (lattice) geometry

The Tessera engine does not work with smooth fields on a manifold; it works with a *simplicial complex* — a triangulation — and discrete differential forms living on it. This section fixes that vocabulary.

Simplicial complexes. A k -*simplex* is the convex hull of $k + 1$ affinely independent points: a vertex (0-simplex), edge (1-simplex), triangle (2-simplex), tetrahedron (3-simplex), pentatope/4-simplex, and so on. A *simplicial complex* $\mathcal{K}$ is a collection of simplices glued so that every face of a simplex is in $\mathcal{K}$ and any two simplices meet in a common face. The proton host in Section 7 is a triangulated closed 4-manifold, a combinatorial S^4 built from pentatopes.

Cochains = discrete differential forms. A k -cochain assigns a number (here a complex number) to every oriented k -simplex. The space of k -cochains is C^k . A 0-cochain is a function on vertices; a 1-cochain is a number per edge (a discrete 1-form); a 2-cochain, a number per triangle; etc. Cochains are the lattice stand-ins for differential forms Ω^k .

Coboundary d and boundary ∂ . The *coboundary* operator $d : C^k \rightarrow C^{k+1}$ is the discrete exterior derivative: $(d\alpha)(\sigma) = \sum_{\tau \in \partial\sigma} [\sigma : \tau] \alpha(\tau)$, summing the cochain over the oriented faces τ of σ with incidence signs $[\sigma : \tau] = \pm 1$. It satisfies $d^2 = 0$ (the discrete Stokes/Poincaré statement). Its adjoint with respect to the inner product defined by the metric (below) is the *codifferential* $\delta : C^k \rightarrow C^{k-1}$, $\delta = d^\dagger$. The *Hodge Laplacian* is

$$\Delta = d\delta + \delta d, \quad (30)$$

and its kernel $\ker \Delta = \mathcal{H}^k$ is the space of *harmonic* k -cochains, whose dimension is the k -th Betti number b_k (a topological invariant counting k -dimensional “holes”). These harmonic cochains are exactly the carried register states of Section 7.

Regge calculus: geometry from edge lengths. Regge calculus encodes the geometry not by a smooth metric tensor but by the set of *squared edge lengths* ℓ_e^2 . Curvature concentrates on the codimension-2 simplices (the “hinges”: triangles in 4D). Around a hinge h the surrounding flat simplices fail to close up by a *deficit angle*

$$\varepsilon_h = 2\pi - \sum_{\sigma \supset h} \theta_{\sigma,h}, \quad (31)$$

where $\theta_{\sigma,h}$ is the dihedral angle that simplex σ subtends at the hinge h . The deficit ε_h is the discrete Ricci scalar curvature smeared on h . The (Euclidean) Regge action is

$$S_{\text{Regge}} = \sum_{\text{hinges } h} A_h \varepsilon_h \quad (A_h = \text{the hinge's volume}), \quad (32)$$

the lattice version of the Einstein–Hilbert action $\int \sqrt{g} R$.

Dual (Voronoi/circumcentric) volumes. The inner product on cochains needs a volume per cell. Each k -simplex σ is paired with its *dual* $(d-k)$ -cell, built from circumcenters (the circumcentric/Voronoi dual). The product of primal and dual volumes, $V_\sigma V_\sigma^*$, sets the Hodge star and hence δ , Δ , and the harmonic inner products. (Concretely the codebase computes these as the “option A” circumcentric dual volumes on each **Simplex**.)

The Lorentzian (complex) action. Spacetime is Lorentzian, not Euclidean: one direction is timelike. On the lattice this means some squared edge lengths are negative ($\ell_e^2 < 0$ for timelike edges), and the dihedral/deficit angles become *boosts* (hyperbolic angles, arccosh) rather than ordinary rotations. The clean bookkeeping (Sorkin; Asante–Dittrich) makes the Regge action *complex*:

$$S \in \mathbb{C}, \quad \text{Re } S \text{ from spacelike hinges,} \quad \text{Im } S \text{ from timelike (boost) hinges.} \quad (33)$$

The imaginary part is real physics, not a numerical artifact: it constrains degrees of freedom (e.g. the overall scale) that the real part leaves loose. Throughout the engine one therefore works with the full complex S and objectives like $\sum_e |\partial_e S|^2$, never $\text{Re } S$ alone. This is background for why the host geometry can carry a definite spin structure at all.

6 The Dirac–Kähler framework: forms as spinors

We now need spin on a lattice. The standard continuum Dirac field needs a frame (vierbein) at each point and is awkward to discretize; the *Dirac–Kähler* (DK) formulation, by contrast, packages spinors as *inhomogeneous differential forms* — exactly the cochains of Section 5 — and so discretizes naturally.

The Dirac–Kähler equation. Collect a form of each degree into one inhomogeneous form $\Phi = \sum_{k=0}^d \phi^{(k)}$, $\phi^{(k)} \in \Omega^k$. Define the operator

$$D := d + \delta. \quad (34)$$

Because $d^2 = \delta^2 = 0$, it squares to the Laplacian,

$$D^2 = (d + \delta)^2 = d\delta + \delta d = \Delta, \quad (35)$$

so D is a “square root of the Laplacian” — a Dirac-type operator. The Dirac–Kähler equation $(D + m)\Phi = 0$ is, after a change of variables, equivalent to several copies of the Dirac equation.

The Kähler–Atiyah isomorphism. The exterior algebra $\Lambda(\mathbb{C}^d)$ of forms, equipped with the Clifford (“geometric”) product, is isomorphic as an algebra to a matrix algebra. For $d = 4$:

$$\boxed{\Lambda(\mathbb{C}^4) \cong M_4(\mathbb{C}), \quad \dim = 2^4 = 16 = \underbrace{4}_{\text{Dirac spinor}} \times \underbrace{4}_{\text{taste}}.} \quad (36)$$

The 16-dimensional space of inhomogeneous forms factorizes as a 4-component Dirac spinor times a 4-fold “taste” (a discretization multiplicity that, in this codebase, plays the role of the per-hole *flavor* index — the “taste” factor of the DK fiber). The spinor factor carries the spin; the taste factor carries flavor. This single isomorphism is why one mesh can carry *both* the spin and the flavor of a quark.

The Clifford generators γ^a . The image of the basis 1-forms under (36) are the gamma matrices γ^a ($a = 0, \dots, 3$), the Clifford action of a 1-cochain. They satisfy

$$\{\gamma^a, \gamma^b\} = 2\eta^{ab}\mathbb{K}, \quad (37)$$

with $\eta^{ab} = \text{diag}(-, +, +, +)$ the (Lorentzian) signature; in the code these are `DiracKahler.gammas`, of dimension `gammaDimension()=4`, and the function `clifford_residual` measures the deviation of (37) from 0 — a certificate that the generators close.

Spin generators and the structural spin- $\frac{1}{2}$. The Lorentz/rotation generators in the Clifford algebra are

$$\boxed{\Sigma_{ij} = \frac{1}{4} [\gamma_i, \gamma_j].} \quad (38)$$

For two distinct *spatial* indices $i \neq j$ (with $\gamma_i^2 = \gamma_j^2 = +\mathbb{K}$ and $\gamma_i\gamma_j = -\gamma_j\gamma_i$), $\Sigma_{ij} = \frac{1}{2}\gamma_i\gamma_j$ and $\Sigma_{ij}^2 = \frac{1}{4}\gamma_i\gamma_j\gamma_i\gamma_j = -\frac{1}{4}\gamma_i^2\gamma_j^2 = -\frac{1}{4}\mathbb{K}$, so Σ_{ij} has eigenvalues $\pm\frac{i}{2}$, i.e. the *spin component* $S_{ij} = -i\Sigma_{ij}$ has eigenvalues $\pm\frac{1}{2}$. Forming the spin Casimir over the three spatial planes,

$$S_{\text{DK}}^2 = \sum_{i < j \in \{1,2,3\}} S_{ij}^2 = 3 \cdot \frac{1}{4}\mathbb{K} = \frac{3}{4}\mathbb{K}. \quad (39)$$

Every Dirac–Kähler field is *structurally spin- $\frac{1}{2}$* : the Casimir is $\frac{3}{4}$, the same fingerprint as a single $\mathbb{C}^2$ (Section 1). A scalar would give 0 and a vector $0, \pm 1$; only a spinor gives the uniform $\pm \frac{1}{2}$ eigenvalues. This is what `dk_spin_readout.py` certifies on the bare mesh.

Structural vs. composite spin — do not confuse them. There are now *two* different “ $\frac{3}{4}$ ”s, and keeping them apart is essential:

- the **structural** spin- $\frac{1}{2}$ of (39) is a property of *one* DK fiber — one quark — guaranteed by the algebra. It is the lattice analog of (7).
- the **composite** total spin of Section 3 is a property of *three* quarks bound together, and equals $\frac{3}{4}$ only for the entangled mixed-symmetry combination (16); for a product it is $\frac{7}{4}$, for the symmetric state $\frac{15}{4}$.

The code reaches the first $\frac{3}{4}$ easily (it is structural); the hard, partly-open problem is the *second* $\frac{3}{4}$.

7 The emergent proton in the Tessera codebase

We can now describe how a proton appears in the engine, and where spin enters.

The host and the optimizer. The computation lives on a closed 4-manifold host (a combinatorial S^4 of pentatopes). A `MultiCobordism` optimizer evolves the geometry by gated local moves (Pachner moves and surgical conings), keeping only moves that improve the stationary-action objective

$$F = \|\nabla S_{\text{Regge}}\|^2 + \Gamma \cdot r_U, \quad (40)$$

which *extremizes* the (complex) action ($\delta S = 0$, never minimizes its magnitude) while r_U pins the carried register. The topology is *emergent*: holes are not hand-placed but grown under relaxation and kept only if they lower F .

The color register as emergent holes. The proton’s three colors appear as three topological holes of the host. On a 4-manifold the relevant homology is degree 3: the color register is the b_3 harmonic sector, read at cochain degree $k = 3$. Removing n disjoint top cells produces $b_3 = n - 1$ independent holes; three holes is the proton’s color register. The carried register state is the harmonic 3-cochain

$$\psi = \text{EigenstateSynthesis}(\text{st}, 3).\text{carriedRepresentative}, \quad (41)$$

a least-action representative over the holes that carries the color representation $[1, \omega, \omega^2]$ with $\omega = e^{2\pi i/3}$ — the discrete realization of the antisymmetric ϵ_{abc} singlet (27). The carried-singlet condition is certified by the register residual $r_{\text{state}} \rightarrow 0$; the singlet $[1, 1, 1]$ is *not* carried (residual ~ 100), which is exactly confinement.

Per-hole spinor extraction. A simplicial complex has no global frame, so spins at different holes are not *a priori* comparable. The extraction (`dk_composite_spin.py`, `dk_spin_wilson.py`) proceeds:

1. **Cell frame.** For each hole's dual cell c , build a frame $F(c)$ — the `cell_frame`: centered principal axes (inertia-tensor eigenvectors) of the cell's *dual-edge* vectors (circumcenter-to-neighboring-circumcenter). Because the dual-edge *lengths* carry the neighbors' sizes, the frame is non-degenerate even when the cell's own vertices are near-symmetric, and it depends only on the vertex *set* (relabel-invariant, gauge-covariant $F \rightarrow RF$).
2. **Spin transport.** The `facet_transport` across a shared facet is the $\text{Spin}(4)$ rotation aligning the two frames, lifted from $SO(4)$ by `rotation_to_spin`; `wilson_line(i, j)` composes these along a BFS dual path — the discrete *spin-connection Wilson line* (open-path holonomy) that makes the comparison frame-independent. A rotation by ϵ produces spinor eigenphases $\pm\epsilon/2$, the Spin double-cover signature checked by `is_double_cover`.
3. **Spinor.** `emergent_spinor` reads the per-hole spinor from ψ by least-squaring the cell's tetrahedral trivector minors against ψ in $F(c)$ and mapping the resulting 3-form into the Clifford algebra, $\Phi = \sum_t \omega_t \gamma_i \gamma_j \gamma_k$, then $s = \Phi \cdot [1, 0, 0, 0]$.

The validation gates (GAUGE invariance, RELABEL invariance) are run *post-hoc* in the test module and `_main_`, never as a loop condition. Reference modules: `examples/cobordism/emergent_optimizer.py`, `examples/cobordism/dk_composite_spin.py`, `examples/cobordism/dk_joint_spin.py`, `examples/cobordism/`

8 The composite-spin readout problem (the heart)

With per-hole spinors in hand, the obvious move is to combine them into the total J^2 and check whether the emergent bound state is a proton ($\frac{3}{4}$) or a Δ ($\frac{15}{4}$). This is where the physics of Section 3 returns with teeth.

The product read and its provable floor. `composite_j2` transports the three holes' spinors to a common frame via the Wilson lines, forms the *product* state $|s_1\rangle \otimes |s_2\rangle \otimes |s_3\rangle$, and reads J^2 . Equivalently, each hole is reduced to a single-qubit Bloch vector $\langle \mathbf{S} \rangle_h$ (Section 1). But a product of three spin- $\frac{1}{2}$ Bloch vectors is exactly the object of (24): its J^2 is the mixture value, floored *above* the proton's $\frac{3}{4}$ and unable to reach it. The code observes precisely this floor: the readout robustly certifies a *three-spin- $\frac{1}{2}$ (baryon)* state — $|\langle \mathbf{S} \rangle| = \frac{1}{2}$ per hole — but J^2 sits at the $n = 3$ mixture ($\sim \frac{9}{4}$ in the joint estimators), never $\frac{3}{4}$. **This is not a bug, a convergence failure, or a pairing problem:** it is the theorem of Section 3 — a product cannot represent an entangled mixed-symmetry state.

The pair-creation build (correcting the two-quark artifact). An earlier attempt read the joint state off a *static*, hand-made color singlet fed only *two* quarks, and found the per-hole flavor index *not* independent — an artifact of underfeeding a state that intrinsically needs three quarks. The correct construction *builds* the proton rather than reading off a hand-made singlet. One co-optimized `MultiCobordism` runs a *simultaneous* pair-creation event,

$$3 \text{ neutral } q\bar{q} \text{ pairs } (\Sigma = 0) \longrightarrow [\text{proton}, \text{antiproton}], \quad (42)$$

with the proton and antiproton legs co-optimized in *one* system (charge conserved end to end). The proton's three quarks are the three emergent holes of the proton output block, accessed read-only via `MultiCobordism.outputs[0].verts` with the relaxed metric copied onto the sub-complex. This build delivers, and validates, two things:

- it **carries the color singlet** ($r_{\text{state}} \rightarrow 0$ on converged seeds), and
- it gives **independent per-hole flavor** (the DK taste/charge is distinguishable across the three quark holes, relative spread $\sim 0.2\text{--}0.5$) — exactly the three-quark structure the two-quark read lacked.

And yet J^2 still sits at the mixture. Even on this correct build, with carried singlet and independent flavor, the composite J^2 remains at the indefinite mixture ($\sim \frac{9}{4}$, not $\frac{3}{4}$):

flavor handling	joint J^2	per-hole J^2
product	≈ 1.5	≈ 1.8
2+1 clustering	≈ 2.2	≈ 2.8

The reason is now familiar: *every available readout reduces each hole to a single-qubit Bloch vector*, so the three-quark spin state it reconstructs is a *product*, and a product provably cannot be the entangled $\frac{3}{4}$ (recall (24): even the product $|uud\rangle$ is $\frac{7}{4}$). The build *creates* the entanglement; the per-hole readout *discards* it. Even charge-clustering the flavors into a 2+1 (u/d) pairing and symmetrizing the uu spins does not help — $|1, 1\rangle |\frac{1}{2}, -\frac{1}{2}\rangle$ is still $\frac{7}{4}$, not the pure $J = \frac{1}{2}$ eigenstate. The validated regression guard (in `test_dk_joint_spin.py`) confirms the *measuring stick* is correct: fed the clean states by hand, the J^2 operator returns $\frac{3}{4}$ (proton eigenstate), $\frac{15}{4}$ (Δ , $|uuu\rangle$), and $\frac{7}{4}$ (product $|uud\rangle$). The instrument is right; the per-hole *sampling* of the emergent state is what loses the entanglement.

9 The total-space spin operator (the resolution being built)

The diagnosis dictates the cure. To recover $\frac{3}{4}$ the spin operator must act on the *whole* carried representative at once — the entangled three-quark state as one object — rather than on a product of per-hole spinors. This is the *total-space spin operator* (scoped in the codebase as the open follow-up; design note `docs/design/total_space_spin_operator_design.md`).

The operator. Let P_h be the projector onto the h -th hole’s degrees of freedom inside the single joint carried field ψ , and let Σ_a (a a spatial rotation plane) be the structural spin generator (38) acting on the DK fiber. The total spin component is $J_a = \sum_h P_h \Sigma_a P_h$, and the Casimir is

$$J^2 = \sum_a \left(\sum_h P_h \Sigma_a P_h \right)^2 = \sum_a \sum_{h, h'} P_h \Sigma_a P_h P_{h'} \Sigma_a P_{h'} \quad (43)$$

acting on the *joint* field. The *diagonal* terms $h = h'$ reproduce the per-hole pieces (the product floor); the crucial *cross-hole* terms $h \neq h'$ are precisely the entanglement carriers that the product read threw away — they are the lattice analog of the swap operators P_{12}, P_{13}, P_{23} in (18) that produced the cancellation $2 - 1 - 1 = 0$ giving $\frac{3}{4}$. Keeping these cross terms is the whole point.

The core difficulty: fibers vs. cells. The 16-dimensional form fiber on which Σ_a acts is defined *point-wise*, but the cochain components of ψ live on simplices of *different degrees* (vertices, edges, triangles, tetrahedra) that do not sit at the same point and do not align point-by-point on a generic simplicial complex. So one cannot simply “apply Σ_a ” to ψ . The assembly needs a map from the cells to the full DK total space — a *Whitney/Kähler–Atiyah* interpolation that lifts the

degree-stratified cochain ψ into the 16-component DK field where Σ_a is defined, applies (43), and projects back. Building this fiber $\leftrightarrow$ cells map is the open engineering task (issue #495, the readout follow-up to the build of #489), and it is the last rung between the validated $\frac{7}{4}/\frac{15}{4}/\frac{3}{4}$ *instrument* and a $\frac{3}{4}$ read *off the emergent proton*.

10 Summary

The numbers, all in one place.

state	J^2	meaning
single spin- $\frac{1}{2}$ (one $\mathbb{C}^2$)	$\frac{3}{4}$	structural spin, Eq.(7)
DK fiber (one quark)	$\frac{3}{4}$	structural, Clifford Σ_{ij} , Eq.(39)
proton, entangled mixed-symmetry $2 uud\rangle - udu\rangle - duu\rangle$	$\frac{3}{4}$	composite $J = \frac{1}{2}$, Eq.(16)
bare product $ uud\rangle$ (per-hole Bloch read)	$\frac{7}{4}$	the mixture floor, Eq.(24)
three-quark mixture floor (current readout)	$\sim \frac{9}{4}$	product of per-hole qubits
Δ , symmetric $ uuu\rangle$	$\frac{15}{4}$	composite $J = \frac{3}{2}$ quartet

Validated vs. open.

- **Validated.** Structural spin- $\frac{1}{2}$ of the DK fiber (`dk_spin_readout.py`); per-hole spinor extraction with gauge/relabel invariance via the dual-edge frame and spin-connection Wilson line (`dk_composite_spin.py`); the emergent color singlet $[1, \omega, \omega^2]$ carried with $r_{\text{state}} \rightarrow 0$; the pair-creation build delivering a carried singlet *and* independent per-hole flavor (`dk_joint_spin.py`); and the J^2 *instrument* itself, which returns $\frac{3}{4}, \frac{7}{4}, \frac{15}{4}$ on hand-fed clean states (regression guard `test_dk_joint_spin.py`).
- **Open.** Reading $\frac{3}{4}$ off the emergent proton, which requires the total-space spin operator (43) acting on the joint carried field — blocked on the fiber $\leftrightarrow$ cells (Whitney/Kähler-Atiyah) assembly. Tracked as issue #495 / #477.

The one sentence to remember. The proton’s spin is an *entangled* three-body quantity; a readout that reduces each quark to its own Bloch vector lands on the product mixture ($\frac{7}{4}$ per ket, $\sim \frac{9}{4}$ for three), and only an operator acting on the whole carried representative at once — keeping the cross-hole terms — can recover $\frac{3}{4}$.

Design notes. `docs/design/joint_proton_spin_findings.md` (the build and the validated measuring stick, #489) and `docs/design/total_space_spin_operator_design.md` (the planned resolution, #495/#477).

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